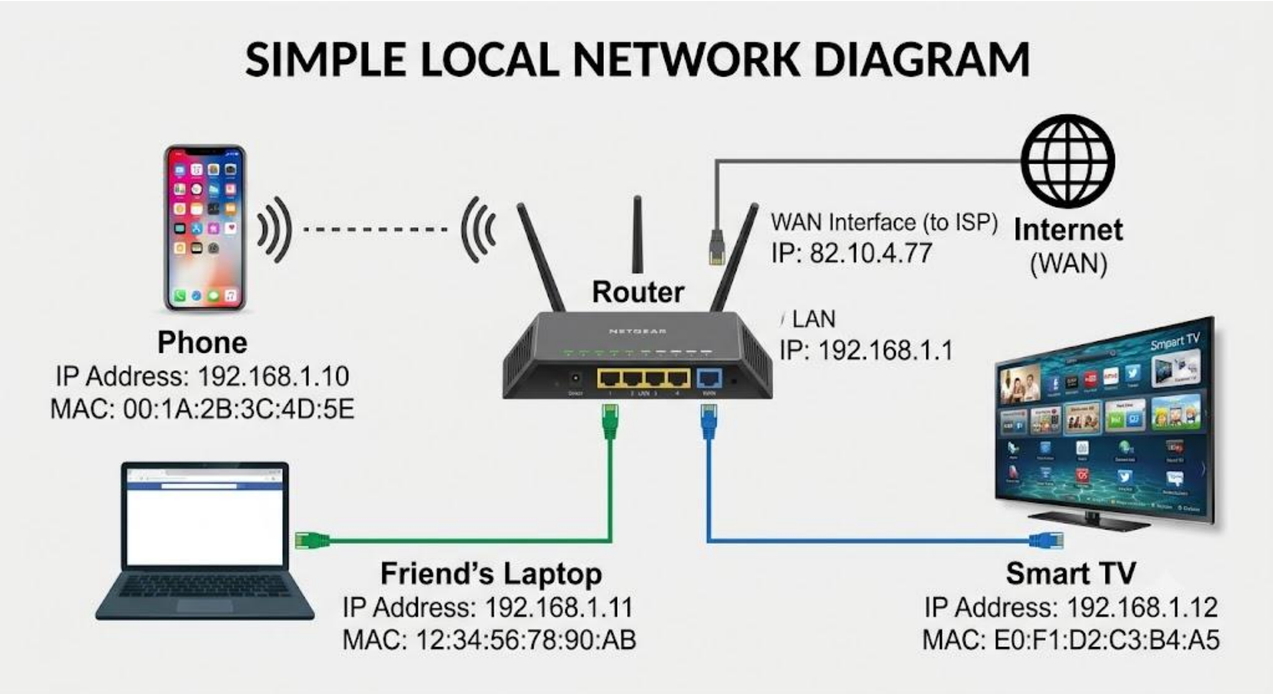
# IP routing and routing protocols

**Task 1 :- Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.**

**ANS :-**



**Task 2 :- Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.**

**ANS :-**

1.Your Laptop :

Looks at your friend's IP, realizes it’s not on your local Wi-Fi, and hands the packet to your home router.

2.Your Home Router :

Replaces your laptop's private IP with its public IP (NAT) and shoots the packet out to your Internet Service Provider.

3.Internet Routers :

A chain of routers act like postal sorting hubs—each looks at the destination IP and passes the packet to next hop closest to your friend.

4.Friend's Router :

Receives the packet, looks up your friend's laptop on its local Wi-Fi (ARP), and hands it over.

5.Friend's Laptop :

Receives and opens the packet.

**Task 3 :- Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.**

**ANS :-**

**Zomato Hub Routing Table :-**

| City / Target Branch | Network Address | Subnet Mask | Next Hop (Where to Send) | Interface (Which Port) | Simple Meaning |
|---|---|---|---|---|---|
| Mumbai HQ | 192.168.10.0 | 255.255.255.0 | Direct (On-link) | eth0 | Local office (no middleman needed) |
| Delhi Hub | 192.168.20.0 | 255.255.255.0 | 10.0.12.2 | eth1 | Pass to Delhi line router |
| Bengaluru Hub | 192.168.30.0 | 255.255.255.0 | 10.0.13.2 | eth2 | Pass to Bengaluru line router |
| Internet / Customer App | 0.0.0.0 | 0.0.0.0 | 203.0.113.1 | wan0 | Everything else goes to the ISP |

**Task 4 :- Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).**

**ANS :-**

**Static Routing (Fixed Route) :-**

*Like a fixed train track.*

- **Pros:**
    1. **Fast & Lightweight :**

       Uses no router power.
    2. **Very Secure :**

       Humans control every path.
    3. **100% Predictable :**

       Always goes the exact same way.

- **Cons :**
    1. **Breaks Easily :**

       If a link dies, traffic stops.
    2. **Hard to Scale :**

       Must be set up by hand.
    3. **Mistake-Prone :**

       Easy to make typos.

- **Real-World Example :**
- A **Zomato billing counter** sending receipts to a local printer over one fixed cable.

**Dynamic Routing (Live GPS / Google Maps) :-**

*Like live traffic rerouting.*

- **Pros:**
    1. **Auto-Fixes :**

        Bypasses broken links automatically.
    2. **Easy Growth :**

        New devices auto-connect.
    3. **Chooses Best Path :**

        Picks the fastest available road.

- **Cons :**
    1. **Uses Resources :**

        Eats router CPU and RAM.
    2. **Slightly Less Secure :**

        Needs extra safety rules.
    3. **Complex :**

        Harder to debug when paths change.

- **Real-World Example :**

    The **Zomato App** instantly switching to a backup server when a primary server crashes.

**Task 5 :- Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.**

**ANS :-**

```
Plaintext

R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 10.1.1.0 0.0.0.255 area 0
R1(config-router)# network 192.168.10.0 0.0.0.255 area 0
R1(config-router)# passive-interface GigabitEthernet0/1
```

router ospf 1 → Turn on OSPF. Starts the routing engine.

router-id 1.1.1.1 → Give it a name. A unique ID so routers recognize each other.

network 10.1.1.0 ... area 0 → Say hello to the neighbor. Starts OSPF on the link connecting R1 to R2.

network 192.168.10.0 ... area 0 → Share local devices. Tells R2 how to reach R1's local office network.

passive-interface GigabitEthernet0/1 → Mute the office port. Shares the office network safely without spamming user PCs.